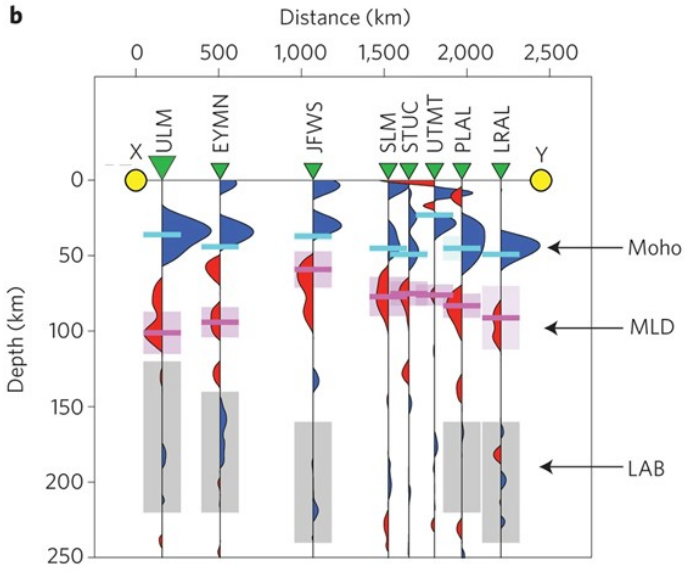
MLD： 低速观测与成因解释

克拉通内部的低速界面

典型深度
80–120 km

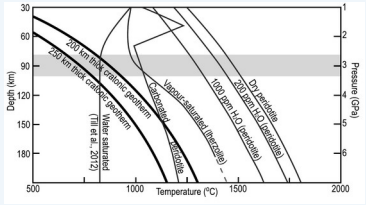
V_S 降幅
约 2–7%



Karato et al. (2015); Fu et al. (2022)

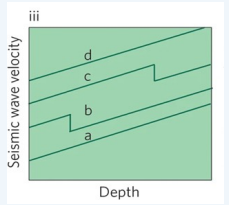
四类成因解释：从物理过程到物质组成

01 部分熔融



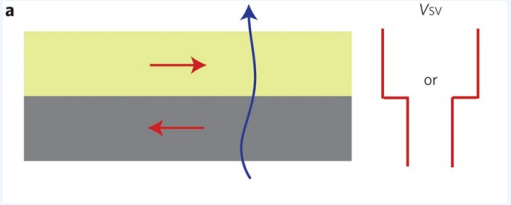
熔体降低有效剪切刚度

02 晶界滑移（EAGBS）



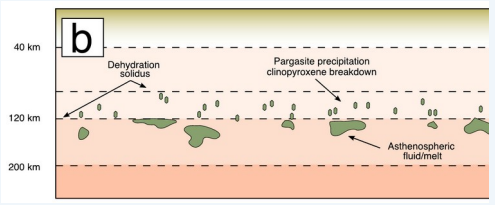
晶界动态松弛降低剪切模量

03 各向异性分层



矿物定向改变产生方向性界面

04 交代矿物富集



矿物组成改变影响全岩弹性

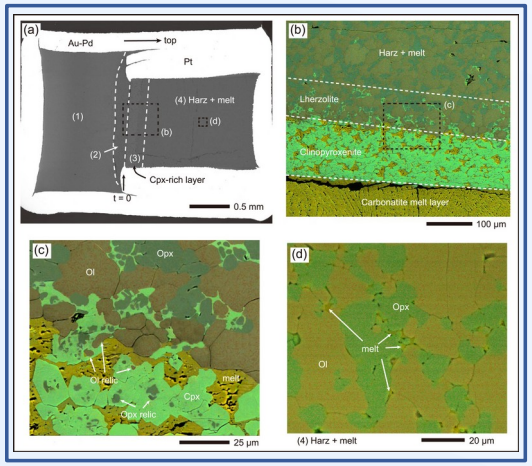
Selway et al. (2015); Karato et al. (2015); Sudholz et al. (2024)

多种机制可致低速，物质组成仍需独立约束

$CaCO_3$：地质依据与高温弹性缺口

物质依据：交代过程＋深部样品

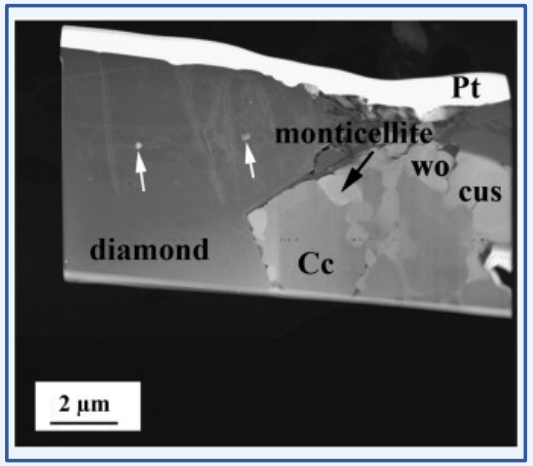
熔体—橄榄岩反应



晶界渗透
并发生交代

Wang et al. (2022), JGR-SE

金刚石包裹体

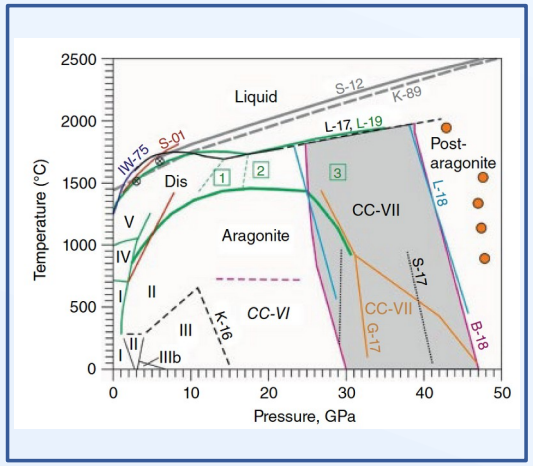


金刚石中保存
方解石及共生矿物

Wirth et al. (2009), EPSL

状态认识：相关系＋高温结构

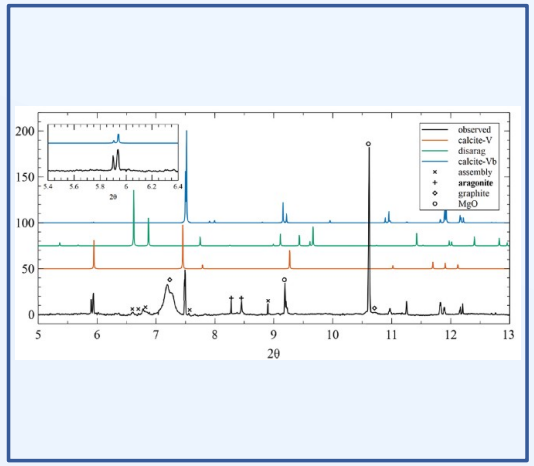
$CaCO_3$ 相关系



相关系复杂
部分相界仍有分歧

Litasov et al. (2020)

高温原位 XRD



高温取向无序
难以淬火保留

Druzhbin et al. (2022), AESCCQ

关键缺口：高温无序固态 $CaCO_3$ 的弹性及全岩波速效应仍待量化